Nuclear Fission and Fusion

Mass defect, binding energy, fission and fusion

Name _____ Class _____ Date _____

CONSTANTS AND DATA

$E = \Delta m\, c^2$ · $c = 2.998 \times 10^8$ m/s · 1 u = 1.66054×10^{-27} kg = 931.5 MeV · 1 MeV = 1.602×10^{-13} J · $N_a = 6.022 \times 10^{23}$ mol^{-1}

Masses (u): proton 1.007276 · neutron 1.008665 · ^{4}He nucleus 4.001506 · ^{2}H atom 2.014102 · ^{3}He atom 3.016029

Z: He 2 · Kr 36 · Sr 38 · Xe 54 · Ba 56 · U 92

Convert mass to kg before using $E = \Delta m\, c^2$. Show units on every quantity.

A. Understanding

1. (a) Define mass defect and binding energy. (b) Use the binding-energy-per-nucleon curve to explain why both fission and fusion release energy.

B. Nuclear equations

2. Complete: (a) ${}^{1}_{0}\mathrm{n} + {}^{235}_{92}\mathrm{U} \longrightarrow {}^{90}_{38}\mathrm{Sr} + {}^{144}_{54}\mathrm{Xe} + \,?$ (b) ${}^{1}_{0}\mathrm{n} + {}^{235}_{92}\mathrm{U} \longrightarrow ? + {}^{92}_{36}\mathrm{Kr} + 3\,{}^{1}_{0}\mathrm{n}$ (c) ${}^{2}_{1}\mathrm{H} + {}^{2}_{1}\mathrm{H} \longrightarrow {}^{3}_{2}\mathrm{He} + \,?$ (d) ${}^{2}_{1}\mathrm{H} + {}^{3}_{1}\mathrm{H} \longrightarrow {}^{4}_{2}\mathrm{He} + \,?$ State whether each is fission or fusion.

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C. Mass and energy

3. A helium-4 nucleus (2 protons, 2 neutrons) has a mass of 4.001506 u. Calculate (a) its mass defect in u and in kg (b) its binding energy in J (c) its binding energy per nucleon in J.

4. How much energy would be released if 1.00 g of matter were converted completely into energy?

D. Fission and reactors

5. Explain the job of (a) the moderator (b) the control rods (c) the coolant in a nuclear reactor. (d) What is meant by critical mass?

6. Give two differences between fission and fusion, and explain why fusion needs temperatures of millions of degrees.

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7. Each fission of uranium-235 releases about 200 MeV. Calculate the energy released (a) per fission, in J (b) per mole of uranium-235 (c) per gram of uranium-235 (235.04 g/mol). (d) Burning methane releases 890 kJ/mol (16.04 g/mol). About how many times more energy per gram does fission release?

E. Fusion and the Sun

8. Calculate the energy, in MeV, released by $^{2}_{1}\mathrm{H} + {}^{2}_{1}\mathrm{H} \longrightarrow {}^{3}_{2}\mathrm{He} + {}^{1}_{0}\mathrm{n}$, using atomic masses from the data box.

9. The Sun radiates 3.83×10^{26} J every second. (a) How much mass does it convert into energy each second? (b) The Sun's mass is 1.989×10^{30} kg. At this rate, how many years would it take to lose 1.00 % of its mass? (1 yr = 3.156×10^{7} s)

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